

MODULE 3 · BONUS WEEK

Agentic AI

From a single LLM answer to a system that takes actions — calling tools in a loop until the task is done.

Lecture 31 + Agentic AI lab



This week

- 1 **LLM call vs. agent** — answering vs. acting
- 2 **Tools** — functions the agent can call
- 3 **The tool loop** — act, observe, decide, repeat
- 4 **A data-ETL agent** — extracting structure autonomously
- 5 **When to use one** — and when not to

What is an agent?

- A traditional LLM call: you send a prompt, you get a response. One shot.
- An **agent**: the LLM decides what to *do* next — call a tool, run code, write a file — and keeps going.
- It loops until the goal is accomplished, not just until it has produced text.

Key idea An agent doesn't just answer — it takes a sequence of actions toward a goal.

Tools

- **Tools** are functions you expose to the agent: search, write a file, run SQL, call an API.
- You describe each tool; the model chooses which to call and with what arguments.
- The agent is only as capable as the tools you give it.

```
def run_sql(query: str) -> str: ...  
def write_file(path: str, text: str) -> str: ...  
tools = [run_sql, write_file]
```

The tool loop

- Agent calls a tool → gets a result → decides the next action → repeats.
- Each turn the model sees what happened and plans the next step.
- Stop conditions matter: finish when done, and cap the number of steps.

Key idea Call a tool, read the result, decide again — that loop is the whole idea of an agent.



A data-ETL agent

- The lab's `data_etl_agent.py` extracts structured data from unstructured text — but as an agent, not a single call.
- It can re-read, retry, and validate its own output across several steps.
- A companion agent writes linear programs from a business description.



When to use an agent

- Good fit: multi-step tasks with tools, retries, and decisions along the way.
- Overkill: a single, well-specified transformation — just call the model once.
- Always keep keys safe in a `.env` file and never commit them.

Lab this week

- Run an agent that loops over tools to accomplish a goal, and trace each tool call → result → next action.
- Set up your API key in a `.env` file (never commit it) and watch the agent work.

In the labs repo: `Module3/WeekN_AgenticAI/data_etl_agent.py` · `Module3/WeekN_AgenticAI/lp_writer_agent.py`



Key takeaways

- An agent acts in a loop; a plain LLM call answers once.
- Tools are the functions the agent can call — it's only as capable as they are.
- The tool loop = act → observe → decide → repeat, with sensible stop conditions.
- Reach for an agent on multi-step tasks; keep it simple otherwise.